

22 2026-03-23 | Week 10 | Lecture 22

22.1 Joint distributions - discrete case

Please read chapter 5.1

We are interested in understanding the properties of **random vectors**, which are vectors whose entries are random variables. We'll start with the case where the random variables are discrete.

Definition 98 (Joint pmf). Let X and Y be two discrete random variables. The **joint pmf** of X and Y is the function

$$p(x, y) = \mathbb{P}[X = x, Y = y],$$

defined for all $x, y \in \mathbb{R}$.

Remark 99.

- It must be the case that $p(x, y) \geq 0$ and $\sum_{x,y} p(x, y) = 1$.
- Let A be any set consisting of pairs of (x, y) values. Then the probability

$$\mathbb{P}[(X, Y) \in A] = \sum_{(x,y) \in A} p(x, y)$$

Example 100. Cars made in a factory experience two kinds of defects: defective joint welds, and and improperly tightened bolts. Let

- X = number of defective welds in a new car
- Y = number of improperly tightened bolts

Past data suggests that the joint pdf of (X, Y) is given by the following table:

		Y			
		0	1	2	3
X	0	.840	.030	.020	.010
	1	.060	.010	.008	.002
	2	.010	.005	.004	.001

The probability that there are no defects in the car is

$$p(0, 0) = \mathbb{P}[X = 0, Y = 0] = .84$$

The probability that there will be exactly one defect is

$$p(0, 1) + p(1, 0) = \mathbb{P}[X = 0, Y = 1] + \mathbb{P}[X = 1, Y = 0] = .06 + .03 = .09.$$

What is the probability that there will be no improperly tightened bolts? This concerns only the variable Y . This can be obtained by summing over all values of x :

$$\begin{aligned}\mathbb{P}[Y = 0] &= \sum_{x=0}^2 \mathbb{P}[X = x, Y = 0] \\ &= p(0, 0) + p(1, 0) + p(2, 0) \\ &= .84 + .06 + .01 \\ &= .91\end{aligned}$$

End of Example 100. $\square$

Given the joint pmf for a two-dimensional discrete random vector (X, Y) , it is easy to derive the individual pmfs for X and Y , and in this context we call them “marginal pmfs”. The manner in which this is done is suggested by the previous example.

Definition 101. If $p(x, y)$ is the pmf of (X, Y) , then the **marginal pmf** of X is

$$p_X(x) = \mathbb{P}[X = x] = \sum_{\text{all } y} p(x, y).$$

Similarly, the **marginal pmf** of Y

$$p_Y(y) = \mathbb{P}[Y = y] = \sum_{\text{all } x} p(x, y).$$

Recall that two events are independent if knowledge that one has occurred gives no information about whether the other has occurred. We can extend this idea to random variables:

Definition 102 (Independence of rvs - discrete case). Let X, Y be discrete random variables with joint pmf $p(x, y)$. We say that X and Y are **independent** if for every pair of (x, y) -values,

$$p(x, y) = p_X(x) \cdot p_Y(y)$$

If this condition isn't satisfied for all (x, y) , then X and Y are said to be **dependent**.

The intuition is this: two random variables are independent if observing one of them doesn't give you any information about the other.

Example 103. In the car factory example, X and Y have with joint pmf is specified by the table

		Y			
		0	1	2	3
X	0	.840	.030	.020	.010
	1	.060	.010	.008	.002
	2	.010	.005	.004	.001

where, for example $p(1, 3) = .002$.

The marginal pmfs are given by the following tables:

x	$p_X(x)$		y	$p_Y(y)$
0	.9	and	0	.91
1	.08		1	.045
2	.02		2	.032
			3	.013

Are X and Y independent? Having computed the pmfs, we can verify that X and Y are **not independent**. This is because

$$p(0, 0) = .84 \neq (.9)(.91) = .819 = p_X(0)p_Y(0).$$

End of Example 103. $\square$

22.2 Joint distributions - continuous case

Based on chapter 5.1 in the textbook – in particular, please read examples 5.5 and 5.14 in the textbook.

The theory for continuous random vectors parallels closely the theory for discrete random variables presented in Section 22.1.

Definition 104. If X, Y are continuous r.v.s, then the **joint pdf** $f(x, y)$ is a function satisfying $f(x, y) \geq 0$ and $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$ such that for any 2-dimensional set A ,

$$\mathbb{P}[(X, Y) \in A] = \int_A \int f(x, y) dx dy$$

The *marginal densities* of X and Y are

$$f_X(x) = \int_{\text{all } y} f(x, y) dy \quad \text{and} \quad f_Y(y) = \int_{\text{all } x} f(x, y) dx$$

We say that X and Y are *independent* if

$$f(x, y) = f_X(x) \cdot f_Y(y).$$

for all $x, y \in \mathbb{R}$.